



Applications of Semi-conductor and Dielectric Materials

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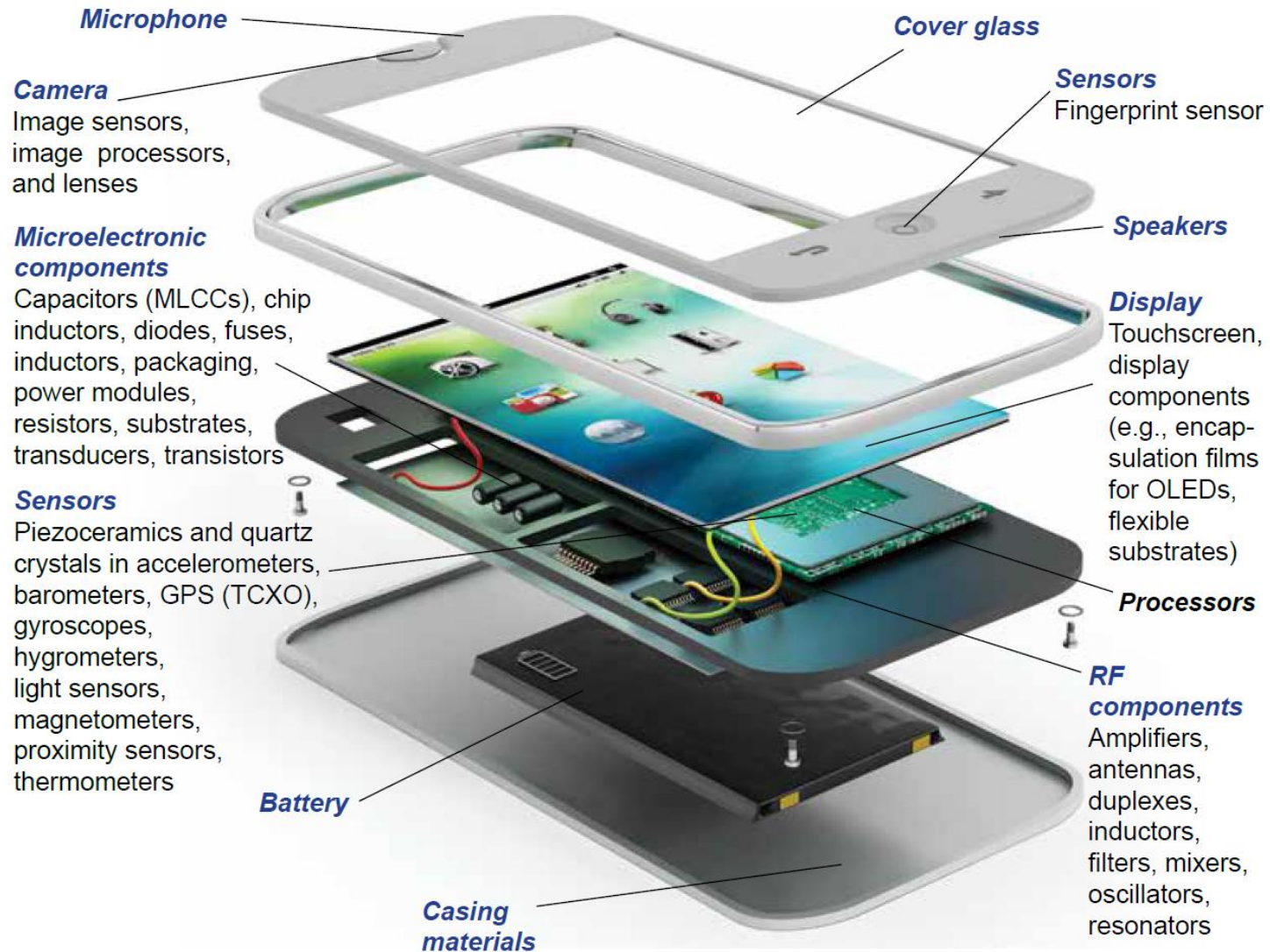
Applications

Applications of **dielectric materials**
in smartphones:

Materials in a **silicon chip**:



Dielectric materials in a phone

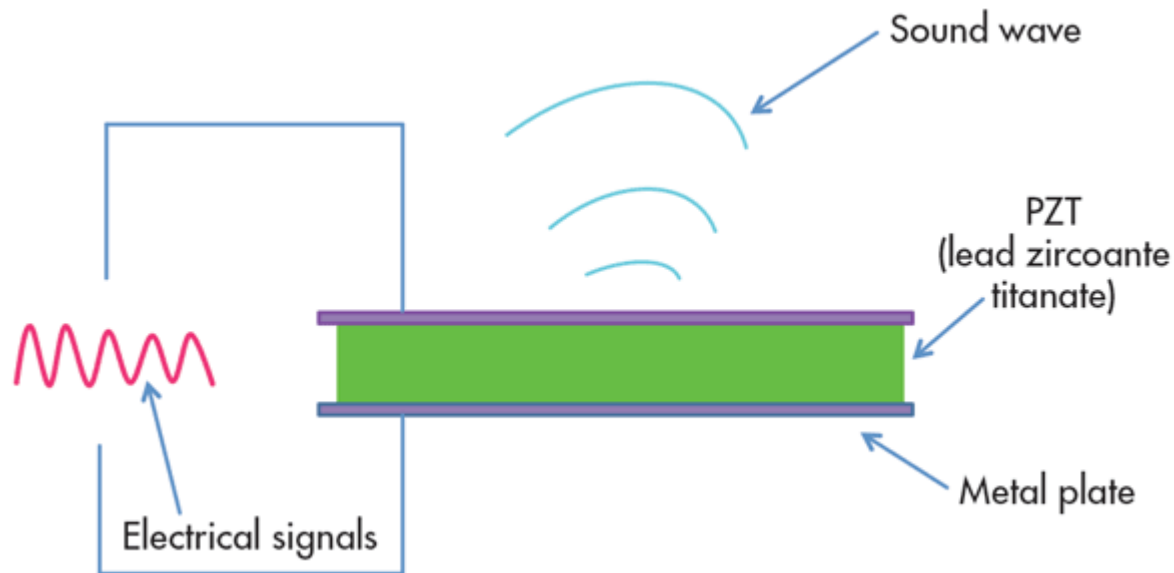


Dielectric materials in a phone

Piezoelectric materials

Used as resonators, actuators and sensors

e.g. speaker, microphone, camera lens movement (for focus)

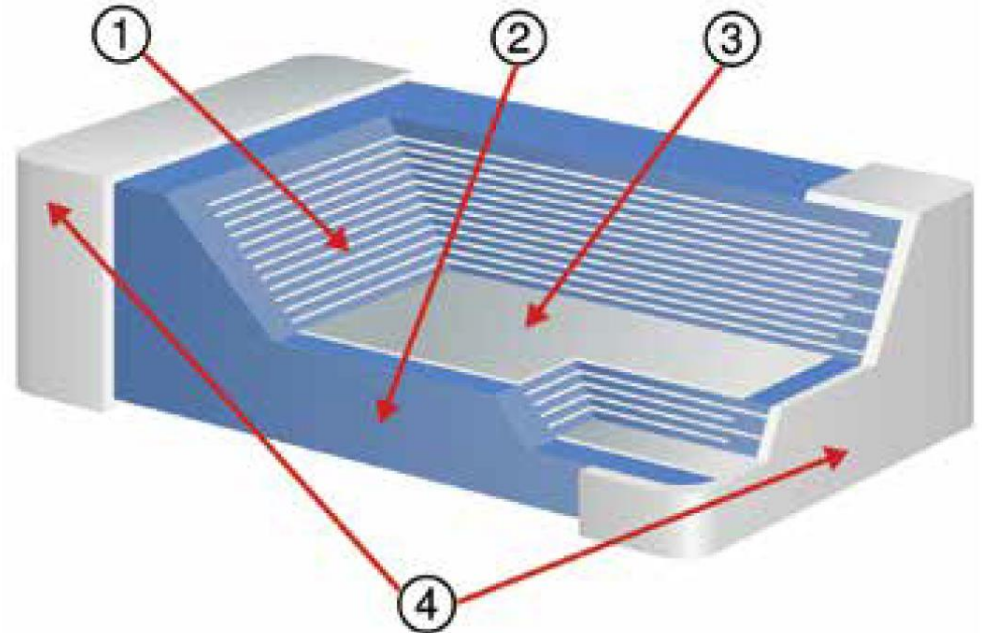


Dielectric materials in a phone

Capacitors

Multilayer Ceramic Capacitor (MLCC)

- ① 100 layers of BaTiO_3
< $1\mu\text{m}$ thickness
- ② Ceramic packaging
- ③ Nickel conductor
- ④ Nickel contacts



Non-volatile memory (Flash)

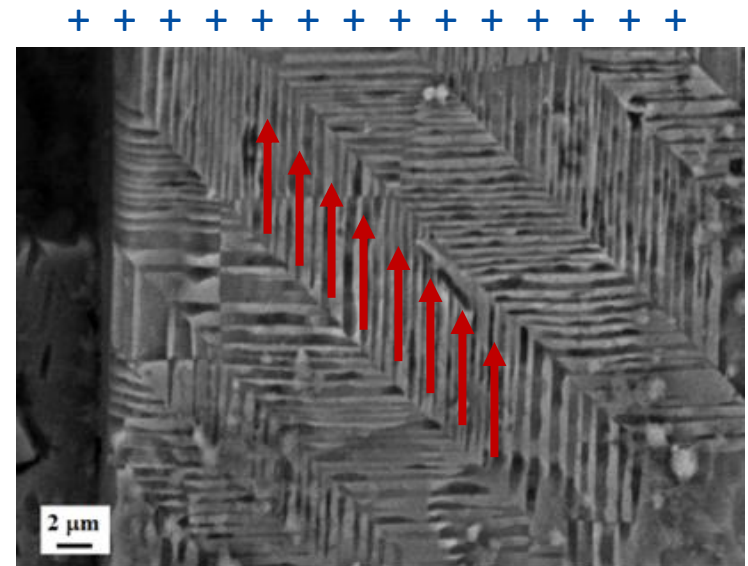
Retained stored information after power supply is removed

Ferroelectric Random Access Memory (FeRAM)

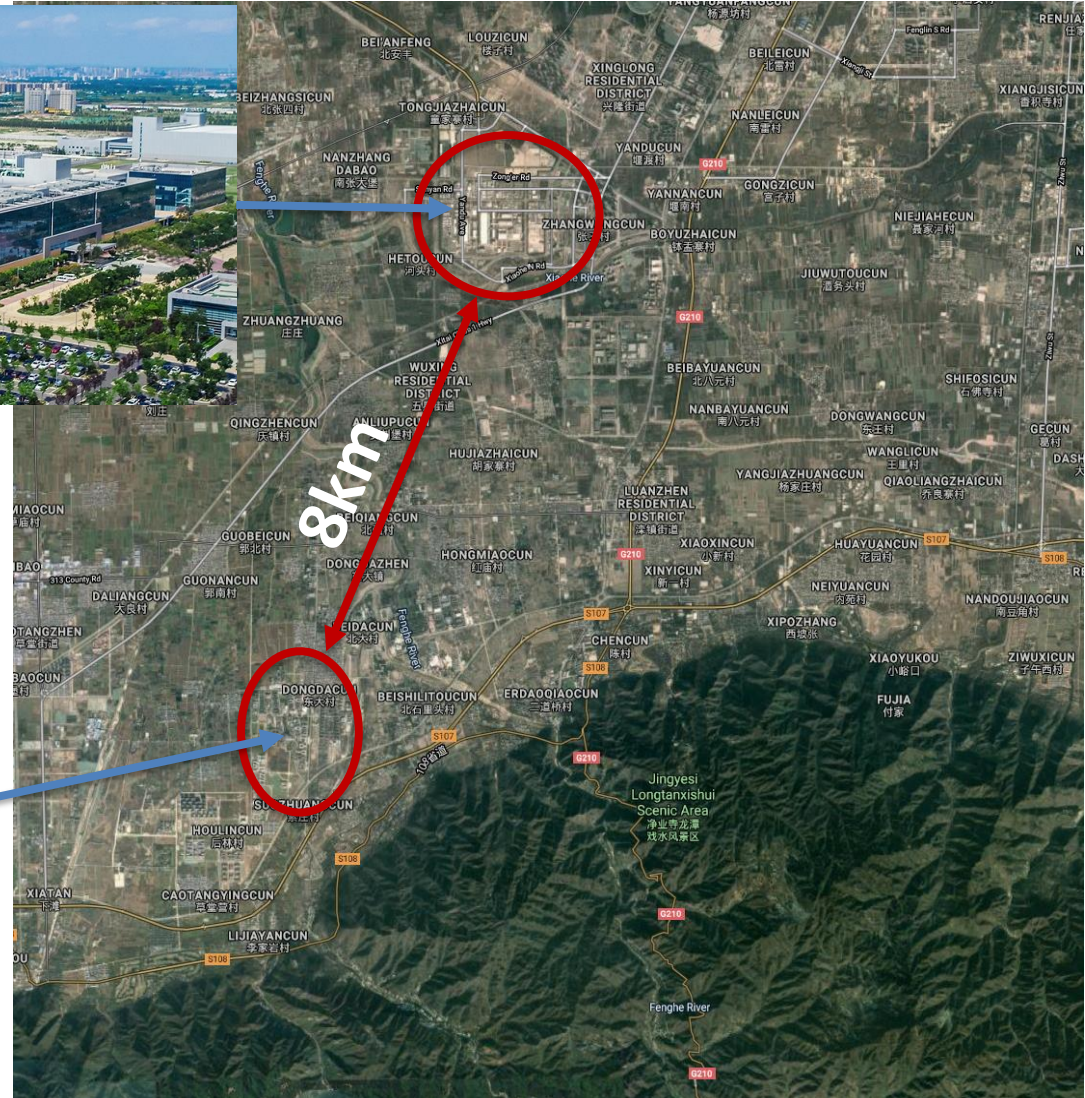
Uses the polarisation of the ferroelectric material (BaTiO_3)

A transistor in series with the RAM
reads the + or – polarisation
of the FeRAM = 0 or 1 in digital code

Application of the electric field can
Switch the polarization from + to -



Silicon chip technology



**Non-volatile memory
made in Xi-an**



Silicon chip technology

Silicon chip microprocessors and memories are used in all our electronic devices

A large number (billions) of devices can be built into an integrated circuit on a small (1cm^2) piece of silicon (chip)

The process begins with a single crystal of silicon up to 300mm in diameter and almost defect free

Cut into thin wafers



Hitachi High-Tech global

The devices keep getting smaller

Why?

More devices = more operations

Smaller distances, l ,

= lower voltage for same electric field strength

$$\mathcal{E} = \frac{V}{l}$$

= less energy needed and less energy loss (heat)

= faster switching

What is the final limit on size?

MOSFET scaling (process nodes)

10 μm – 1971

6 μm – 1974

3 μm – 1977

1.5 μm – 1981

1 μm – 1984

800 nm – 1987

600 nm – 1990

350 nm – 1993

250 nm – 1996

180 nm – 1999

130 nm – 2001

90 nm – 2003

65 nm – 2005

45 nm – 2007

32 nm – 2009

22 nm – 2012

14 nm – 2014

10 nm – 2016

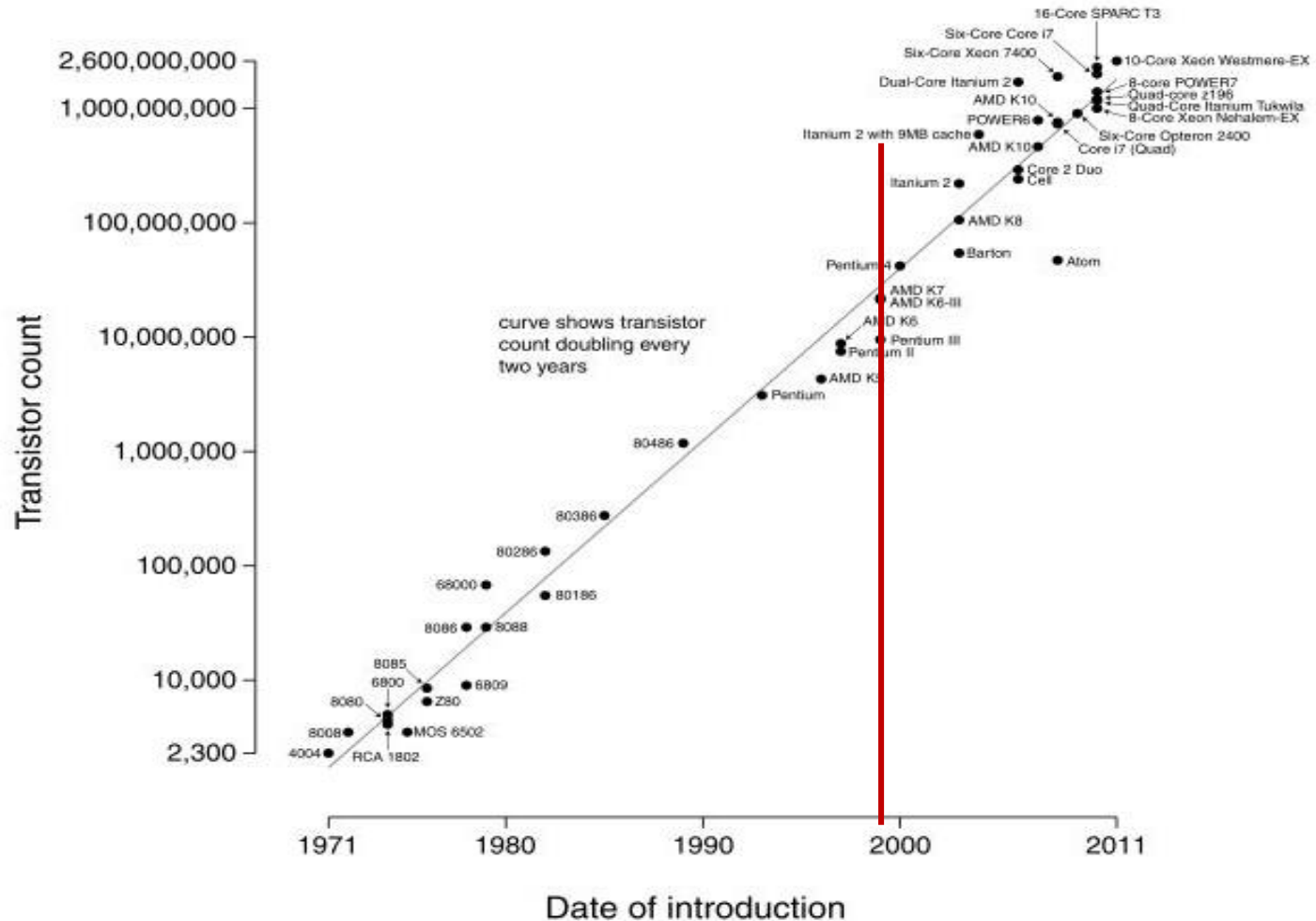
7 nm – 2018

5 nm – 2020

Silicon chip technology



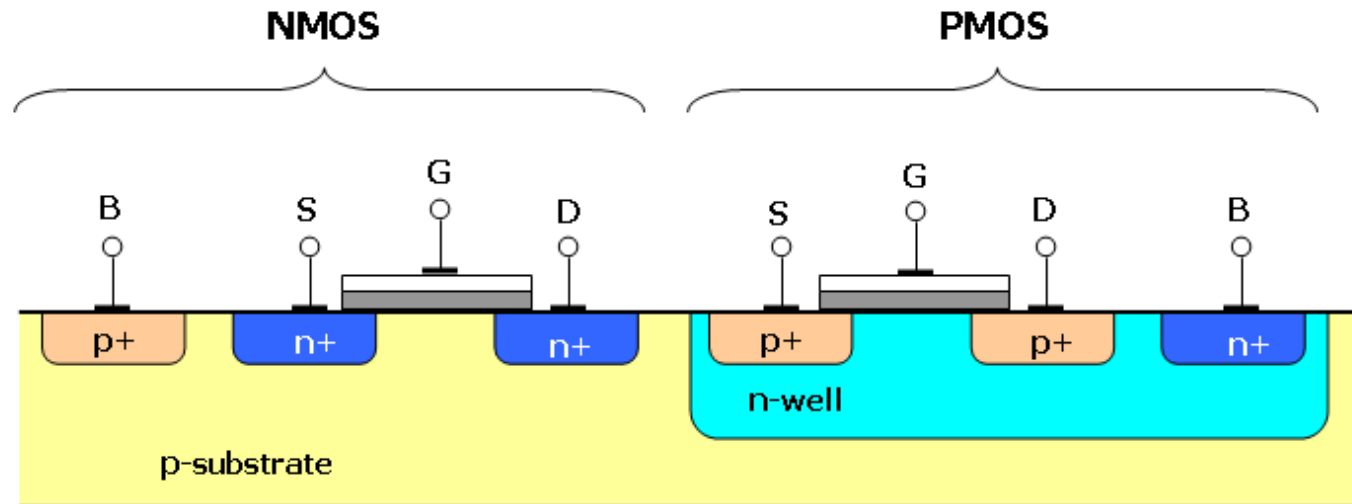
Moore's Law – the number of transistors in an integrated circuit doubles every 2 years



Silicon chip technology

CMOS – complementary metal oxide semi-conductor

Uses complementary pairs of p-type and n-type MOS-FETs



Materials used:

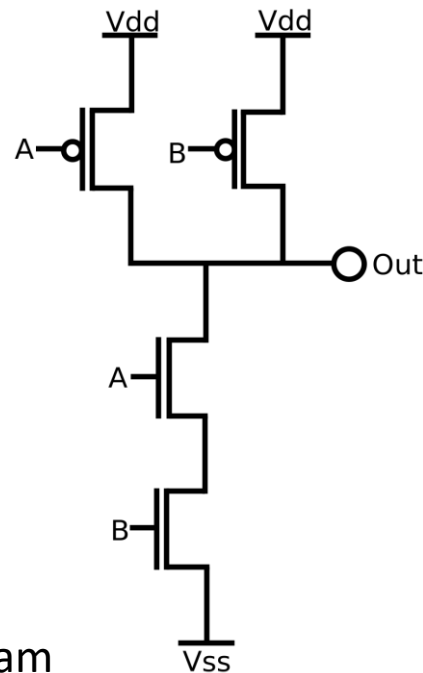
- p-type silicon wafer – single crystal, polycrystalline silicon (extra semi-con)
- n-type and p-type dopants (impurities) by vapour deposition and diffusion
- High k dielectrics to separate the gate from the devices
- metal connections

Silicon chip technology

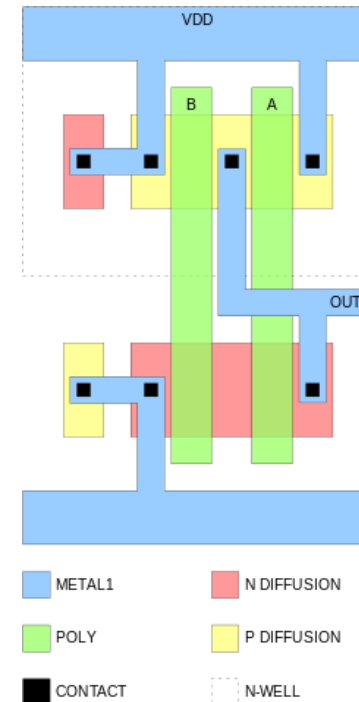
CMOS – complementary metal oxide semi-conductor

Uses complementary pairs of p-type and n-type MOS-FETs

Example NAND gate ('NOT-AND') logic circuit



Circuit diagram

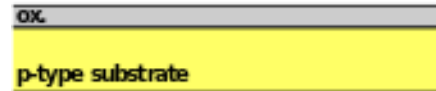


Physical layout

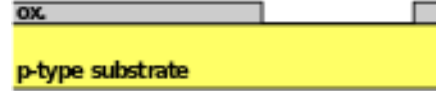
Silicon chip technology

NAND gate manufacture steps

1. Grow field oxide



2. Etch oxide for pMOSFET



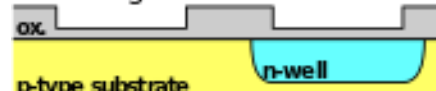
3. Diffuse n-well



4. Etch oxide for nMOSFET



5. Grow gate oxide



6. Deposit polysilicon

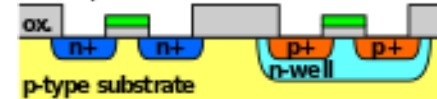


7. Etch polysilicon and oxide



Extreme UV photolithography used to pattern many millions of devices on the chip
Photoresist materials are polymers

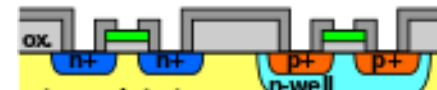
8. Implant sources and drains



9. Grow nitride



10. Etch nitride



11. Deposit metal



12. Etch metal



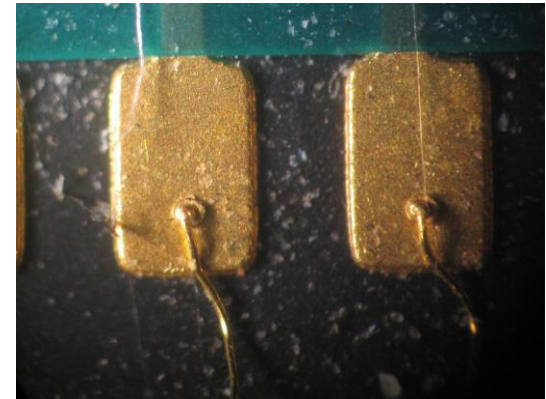
Other materials problems

Metallization:

The metal connections to the devices (conductors) were originally made of **aluminium**. As lines became smaller the current density became larger and the lines failed by **electromigration** (movement of metal with the current). Now **copper** is used (higher melting temperature)

Wire bonding:

Joining the integrated circuit to the rest of the electronics is done with **gold** wires, 15 μm in diameter, bonded to gold pads by combination of pressure, heat and ultrasonic vibration



Low k dielectrics:

Low dielectric constant materials are needed as insulators for the circuit with inducing (unwanted) parasitic capacitance. E.g polymers such as polyimide

Electrical properties:

Technology uses electrical properties for a range of applications

- Conductivity
- Semi-conduction
- Dielectric behaviour

- Capacitance
- Polarisation

Learning Outcomes

What can you do now?

- Know the difference between electrical properties
- Conduction of electrons in solids and the relation to the electronic band structure of solids
- Intrinsic and extrinsic semi-conductor behaviour
- Dielectric materials and polarisation behaviour
- Select materials for their electrical applications